

JOSHUA ALCOBIA GOMES

London • jgZalcobia@gmail.com • +41 76 305 71 17 • [LinkedIn Profile](#)

EDUCATION

University College London - MEng Mechanical Engineering
2nd Year – Predicted 1st Class Honours (83%)

Sept 2024 – June 2028
London, UK

International School Basel (ISB) - International Baccalaureate

Result: HL Maths AA 7, HL Physics 7, HL Economics 7, SL Chemistry 7, SL German B 7, SL English A L&L 6

Aug 2022 – Jun 2024
Basel, Switzerland

EXPERIENCE

Novartis

Scientific Modelling Intern

Aug 2025 – Sept 2025
Basel, Switzerland

- Built Streamlit applications with dropdown menus for dwell time analysis and screw customisation, eliminating manual data searching for the department.
- Developed **Python programs for Loss on Drying calculations** from sensor data, solving **complex mass balance equations** to determine moisture content and drying efficiency in pharmaceutical manufacturing processes.
- Created **machine learning models** to predict powder flow characteristics using **LazyClassifier** for model selection and **Grid Search, Lasso regularisation, and Bayesian optimisation** for hyperparameter tuning, **saving CHF 40,000 per batch of APIs**.
- Developed an AI Project Summariser using **automated workflows and LLMs** to gather data from external and internal databases, **saving weeks and reducing research time from 50 to 15 minutes**.
- Designed tablets using SolidWorks** for COMSOL HPC simulation based on provided technical drawings, **saving thousands per batch** through predictive modelling.

University College London Racing | Rocket

National Rocketry Championship Team Lead, Aerostructures and Simulations Engineer, Sponsorship Team Member

Oct 2024 – Present
London, UK

- Led an 8-engineer team with a £1,000 budget to **8th place** out of 50 at the National Rocketry Championship.
- Validated **CFD** simulations in **ANSYS Fluent** with drag coefficient data, with 97% accuracy.
- Created Gantt charts** and learned **Atlassian's Jira** project management software to ensure workload was properly divided and maintained a structured order of operations throughout the project lifecycle.
- Wrote comprehensive risk assessments** to ensure proper safety protocols during rocket construction phases and on launch day, identifying potential hazards and implementing mitigation strategies.
- Designed the rocket's aerostructure in **Fusion 360**, applying **aerodynamic principles** to increase apogee by 203m.
- Secured project funding through targeted sponsorship outreach to industry partners.

Institution of Mechanical Engineers Design Challenge

Design & Manufacturing Engineer

Feb 2025 – Present
London, UK

- Achieved runner-up in the Greater London Regional Final, securing **National Final qualification** as one of only two teams to achieve perfect accuracy.
- Contributed to all project stages from planning and design to testing, winning both the **Design Award** and **Challenge Competition**.
- Designed electrical circuits with resistors**, considering **voltage, power, and current limitations** to ensure all components operated within safe parameters and met design specifications.
- Used **Fusion 360's** Generative Design to optimise a structure, **reducing mass by 115.87g (60%)** while maintaining a safety factor of 2.
- Produced a comprehensive **Bill of Materials** with data visualisation to identify cost drivers for decision-making.

CERN Beamline for Schools

Physics Researcher

Feb 2022 – Sept 2023
Basel, Switzerland

- Co-developed a comprehensive experiment proposal for CERN with a team of 11, investigating the relationship between particle beamline energy and positron emissions.
- Researched and designed the experimental testing set-up, including appropriate detectors and beamlines to produce desired data.
- Deepened my understanding of particle physics while showcasing our team's ability to work on complex scientific challenges.

ISB Student Council

President (22-23), Grade Representative (23)

Sept 2022 – Apr 2024
Basel, Switzerland

- Led planning and execution of multiple events, each **attracting over 500 attendees**, including growing Prom attendance from **108 to 156 participants (44% increase)**.
- Coordinated multiple teams, ensuring operations and collaboration with a **diverse range of vendors**.
- Managed a budget of CHF 15,000 and raised over CHF 3,000 through strategic fundraising initiatives.

Self-Employed Tutoring

Tutor

Aug 2023 – Present
International

- Provide tutoring for 6 Middle and High School students in IB, A Levels, MYP, and GCSEs, **improving average student grades from IB 4 to IB 6**.
- Mentor students in **problem-solving and time management** while adapting teaching techniques to individual learning styles.
- Guide mentees in creating effective **time management strategies** to enhance academic performance and balance workloads.
- Track student progress through assessments, offering constructive feedback to promote continuous improvement.

ISB Robotics Summer Camp

Mentor

Jun 2022 – Jun 2022
Basel, Switzerland

- Worked alongside primary school students (7-11-year-olds), developing their **problem-solving** skills.

- **Taught VEXcode (block) programming** through intuitive daily tasks using VEX hardware and software.
- Encouraged students to solve problems using various methods, catering the tasks to various skill levels and ages.
- Explored different programming techniques by applying them to **real-world applications**, encouraging students to invest in STEM.

VEX Robotics

Aug 2021 – Mar 2024
Basel, Switzerland

Design & Build Robotic Engineer

- Collaborated in a team of 6 to design, manufacture, and iterate a high-performing competitive robot.
- Contributed to team success, achieving runner-up in the National Competition and winning the Design Book Award.
- Designed an efficient launching system to maximise competition scores.

InvestIn - Introduction to Engineering

Jul 2023 – Aug 2023
London, UK

Engineer

- Worked collaboratively on mini projects regarding tensile and shear strength of different materials in bridges.
- Designed and tested aerofoils to determine the optimal NACA numbers for the maximum stall angle.
- Investigated various flight mechanics and applied our knowledge to gliders and a flight simulation.

PROJECTS

University College London Mechathon 2025

Jan 2025 – Jan 2025
London, UK

- Won the competition (team of 6) with an **efficiency of 29%**, a voltage of 3.3V, with a 76mm radius turbine.
- Used QBlade software to analyse the **lift-to-drag** ratio of different aerofoils to determine the highest ratio and what angle of attack this occurs at; our analysis was **accurate to 95.8%**.
- Developed manufacturing methods, including **milling, 3D printing, and laser cutting**, considering **material properties** to determine manufacturing methods for different components.

The Behaviour of Internal Pressure of Thin-Walled Vessels During Collision

Feb 2023 – Nov 2023
Basel, Switzerland

- Conducted **theoretical research and mathematical modelling** on the relationship between internal pressure and impact force.
- Wrote 4000-word research paper on optimal pressure for impact reduction and applications to safety systems.

Development of a Kitchen Robot for Automated Fruit and Vegetable Cutting

Aug 2021 – Jan 2022
Basel, Switzerland

- Designed, built, and programmed an automated kitchen robot using **Arduino and C++**.
- Modelled and **optimised components in CAD** for functionality and durability.
- Assembled and tested prototypes, **iteratively refining mechanical systems and control algorithms**.
- Strengthened expertise in **robotics, electronics, and mechanical systems**, with a focus on automation in practical applications.
- Documented the project through detailed **technical writing**, including challenges, solutions, and performance outcomes.

SKILLS

CAD/CAM/Modelling Packages: Autodesk Fusion 360, SolidWorks, Onshape, Ansys Fluent, Ansys Mechanical, COMSOL, OpenRocket, QBlade

Programming Languages: Python, MATLAB, C++

Machinery Proficiency: 3D Printers, Sanding Belts, Pillar Drills, Hand Tools, Soldering, Laser cutting

Languages: English (Fluent), German (C1), Mandarin (Beginner)

AWARDS

International Baccalaureate Physics Award for Excellence in Physics

International School of Basel Leadership Award, demonstrating authentic leadership and collaboration for the benefit of others and striving for inclusion within society.

International School Basel Design Award for project management, collaboration, and skill development

VEX Robotics National Championship Runner and **Best Design Award**

Model United Nations Best Delegate (2022) and **Best Chair (2023)**

EXTRA-CURRICULAR ACTIVITIES

- **Track and Field** – Competed nationally, finishing in the top 8 at 2 U20 national championships. Consistently training every day and currently competing with UCL RAX and Highgate Harriers. Trained in Switzerland and the United Kingdom.
- **Model United Nations** – Served as a chair for two years, moderating debates among delegates. Attended THIMUN a prestigious international MUN conference, passing a resolution in a council of hundreds.
- **Coaching** – Worked alongside coaches to tailor training programmes to students. Additionally, generated the training programme for the school's cross-country team for two years, guiding the team to two SCIS championships.

GALLUP CLIFTON STRENGTHS ASSESSMENT

1. **Learner** – The process of discovering the path to a solution is more important than the result
2. **Achiever** – The constant need to achieve tangible change at the end of a project
3. **Realtor** – Continually building professional relationships to develop strategies to solve problems
4. **Focus** – Successful ability to set goals and work continuously toward these goals
5. **Discipline** – Focused on deadlines and planning for precision in working environments